

Handout 6.1: Farm to Plate Path Cards

Print one food set per team and cut the cards apart. Each set has ten step cards and two decoy cards. Print page 4, the Path Recording Sheet, one per student. Print page 5, the blank set, only for the grade 8 stretch. For grade 6, pull the two cards marked [**pull for grade 6**] and use one decoy instead of two.

The time and cost numbers on these cards are teaching estimates, rounded. They are close enough to argue with and not exact.

The ten steps, in order, every food: **grow or raise, harvest, process, package, transport, store, sell, cook, eat, leftovers and waste.**

Set 1: A HALF GALLON OF MILK

Card	Text on the card
GROW OR RAISE	A dairy farm keeps about 120 cows. Each one eats about 100 pounds of feed a day and drinks about a bathtub of water. Job: dairy farmer, herd manager. Time: every day, all year. Cost share (the part of the price that pays for this step): about 30 cents of the final price.
HARVEST	Cows are milked twice a day by machine into a refrigerated tank. The milk is cooled to 38 degrees within minutes. Job: milking technician. Time: 2 to 3 hours per milking. Cost share: included above. Card note: cold the whole way starts here.
PROCESS	A tanker truck brings the milk to a plant. It is tested and pasteurized (heated to kill germs, then cooled fast). Then it is split into whole, 2 percent, and skim. Job: plant operator, quality lab technician. Time: same day. Cost share: about 45 cents. [pull for grade 6]
PACKAGE	The milk is filled into cartons or jugs, sealed, and date stamped. Job: filler line operator. Time: minutes per carton. Cost share: about 30 cents.
TRANSPORT	A refrigerated truck carries pallets to a distribution warehouse. Job: truck driver, loader. Time: 1 day. Distance: usually under 200 miles. Cost share: about 35 cents.
STORE	The warehouse holds it in a cooler at 36 to 38 degrees. Job: warehouse picker, refrigeration mechanic. Time: 1 to 3 days. Cost share: about 15 cents.
SELL	A second truck brings it to the store. A clerk rotates the shelf so the oldest date sells first. Job: delivery driver, stock clerk, cashier. Time: 1 to 5 days on the shelf. Cost share: about 70 cents.
COOK	Most milk is not cooked. It is poured. Some is heated for cocoa or used in a recipe. Job: you. Time: 0 to 5 minutes. Cost share: 0.

EAT	A half gallon is about 8 servings. Job: you and whoever you live with. Time: 3 to 7 days.
LEFTOVERS AND WASTE	Milk that passes its date gets poured out. About 1 glass in 5 of all milk bought in this country is never drunk. The carton is recyclable in most towns. Job: you. [pull for grade 6]
DECOY 1	The farmer sets the price consumers pay at the store.
DECOY 2	Milk is frozen for shipping and thawed at the store.

Total time from cow to your glass: about 6 to 12 days. Number of people paid along the way: about 19.

Set 2: A LOAF OF SANDWICH BREAD

Card	Text on the card
GROW OR RAISE	A farm in the Midwest plants winter wheat in the fall and cuts it the next summer. Job: grain farmer. Time: 9 months in the ground. Cost share: about 10 cents of the final price.
HARVEST	A combine cuts the wheat and separates the grain from the stalk in one pass. The grain goes to a silo. Job: combine operator, elevator worker. Time: 2 to 3 weeks for a whole farm. Cost share: included above.
PROCESS	A mill grinds the wheat into flour and sifts it. A bakery mixes flour, water, yeast, sugar, salt, and oil, lets the dough rise, and bakes it. Job: miller, baker, mixer operator. Time: 1 day at the bakery. Cost share: about 85 cents.
PACKAGE	Loaves are cooled, sliced, bagged, and tagged with a color-coded closure for the day of the week. Job: packaging line worker. Time: about 1 hour. Cost share: about 20 cents. [pull for grade 6]
TRANSPORT	A route driver delivers straight from the bakery to stores, often overnight. Job: route driver. Time: hours. Distance: usually under 150 miles. Cost share: about 30 cents.
STORE	Bread is not refrigerated. It sits at room temperature and goes stale in days. Job: stock clerk. Time: 1 to 3 days on the shelf. Cost share: about 10 cents.
SELL	The store shelves it and the route driver pulls anything past date on the next trip. Job: stock clerk, cashier. Time: 1 to 5 days. Cost share: about 90 cents.
COOK	Some of the loaf is toasted, some is eaten as is, some becomes a sandwich. Job: you. Time: 0 to 3 minutes. Cost share: 0.
EAT	A loaf is about 20 slices. Job: you and whoever you live with. Time: 4 to 10 days.
LEFTOVERS AND WASTE	The heels and the stale end are the most

	thrown-out part of a loaf. Stale bread can be croutons, crumbs, or French toast. The bag is a plastic film, which most curbside programs will not take. Job: you. [pull for grade 6]
DECOY 1	Bread has to be refrigerated the whole way from the bakery to keep it fresh.
DECOY 2	Wheat is ground into flour at the store's own bakery.

Total time from field to your sandwich: about 9 to 10 months, of which 3 days is the part after the farm. Number of people paid along the way: about 15.

Set 3: A BANANA

Card	Text on the card
GROW OR RAISE	Bananas grow on large farms in Central and South America. A plant takes 9 to 12 months to produce one bunch. Job: farm worker, field supervisor. Time: 9 to 12 months. Cost share: about 3 cents of the price of one banana.
HARVEST	The bunch is cut down green by hand while it is still hard. Bananas are never picked ripe, because a ripe banana cannot travel. Job: harvest crew, packing shed worker. Time: one day. Card note: this is the whole trick of the banana.
PROCESS	At a packing shed the bunch is washed, split into hands, checked, and stickered. There is no cooking or grinding step. Job: packing shed worker, quality inspector. Time: same day. Cost share: about 2 cents. [pull for grade 6]
PACKAGE	Hands go into boxes with a plastic liner that slows ripening. Job: box packer. Time: hours. Cost share: about 2 cents.
TRANSPORT	Boxes ride a refrigerated ship at about 56 degrees for 7 to 12 days, then a truck. Distance: about 2,000 miles or more. Job: dock worker, ship crew, truck driver, customs broker. Time: 1 to 2 weeks. Cost share: about 5 cents.
STORE	At a ripening room in the United States the bananas are exposed to ethylene gas, which starts the ripening on a schedule the store chooses. Job: ripening room technician. Time: 4 to 8 days. Cost share: about 2 cents.
SELL	The store puts them out at the ripeness that sells. Job: produce clerk, cashier. Time: 1 to 4 days. Cost share: about 11 cents.
COOK	Most bananas are not cooked. Some go into bread, pudding, or a smoothie. Job: you. Time: 0 minutes. Cost share: 0.
EAT	One banana is one serving. Job: you. Time: minutes.

LEFTOVERS AND WASTE	The peel is about 35 percent of the weight and is compostable. Bananas that get too ripe on the shelf are pulled. Too-ripe bananas at home are the best ones for banana bread. Job: you. [pull for grade 6]
DECOY 1	Bananas are picked ripe and shipped fast so they arrive yellow.
DECOY 2	Most bananas eaten in New York are grown in Florida.

Total time from plant to your hand: about 10 to 13 months, of which about 3 weeks is the trip. Distance: about 2,000 miles. Number of people paid along the way: about 22.

Set 4: A FROZEN CHICKEN NUGGET

Card	Text on the card
GROW OR RAISE	Chickens raised for meat reach market weight in about 6 to 7 weeks. A single barn may hold 20,000 birds. Job: poultry grower, feed supplier, veterinarian. Time: 6 to 7 weeks. Cost share: about 20 percent of the final price.
HARVEST	Birds are transported to a processing plant, where they are slaughtered and cut into parts under federal inspection. Job: catch crew, plant worker, USDA inspector. Time: 1 day. Card note: this is the step most people never think about.
PROCESS	Trim and breast meat are ground and mixed with water, salt, and binders (things that hold it together). The mix is formed into nugget shapes and coated in batter and breading. It is fried for a few seconds to set the coating, then fully cooked. Job: formulation technician, line operator, food scientist. Time: 1 day. Cost share: about 30 percent. [pull for grade 6]
PACKAGE	Nuggets are blast frozen, weighed, bagged, sealed, metal detected, and boxed. Job: packaging operator, quality technician. Time: hours. Cost share: about 10 percent.
TRANSPORT	A frozen truck at 0 degrees carries pallets to a frozen warehouse and then to stores. Job: refrigerated truck driver. Time: 1 to 5 days. Distance: 300 to 1,500 miles. Cost share: about 10 percent. Card note: cold the whole way, and colder than the milk.
STORE	Frozen warehouses hold nuggets for weeks or months. Job: warehouse worker, refrigeration mechanic. Time: up to 9 months. Cost share: about 5 percent.
SELL	The store freezer case holds them until sold. Job: frozen foods clerk, cashier. Time: days to weeks. Cost share: about 25 percent.
COOK	Baked in an oven at 400 degrees for about 12 minutes, or air fried, or microwaved.

	They are already fully cooked, so this step is reheating. Job: you. Time: 12 minutes. Cost share: 0.
EAT	A serving is about 5 nuggets. Job: you and whoever you live with. Time: minutes.
LEFTOVERS AND WASTE	Cooked nuggets left out more than two hours are not safe to keep (the danger zone rule from Unit 1). The bag is a plastic film. Job: you. [pull for grade 6]
DECOY 1	Chicken nuggets are raw when you buy them and cooking them at home is what makes them safe.
DECOY 2	Nuggets are made from one whole piece of breast meat per nugget.

Total time from chick to your plate: about 2 to 11 months depending on how long it sat frozen. Number of people paid along the way: about 25, the most of the four.

Path Recording Sheet

Name: _____ Date: _____ Period: _____

My team's food: _____

Part 1. The ten steps

#	Step name	One job that happens here	Time this step takes	Cost or price share
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Part 2. Decoys

The two cards that did not belong were:

1. _____ Why it is wrong: _____
2. _____ Why it is wrong: _____

Part 3. Time and cost

Go to your wall path. Put a **T** on the step that takes the longest. Put a **C** on the step that adds the most to the price.

Longest step: _____ How long: _____

Biggest price step: _____ How much: _____

Does the farmer get the biggest share of the price? Circle: **yes** / **no**

One sentence on why: _____

Part 4. Gallery walk

Walk the other three paths. Fill this in.

Food	Number of steps that need cold or freezing	Total time from start to plate	One thing that surprised me
Milk			
Bread			
Banana			
Chicken nugget			

Which food has the shortest and simplest chain? _____

Evidence 1: _____

Evidence 2: _____

Exit card

The step I never thought about is _____, and the person who does it is a _____.

Grade 8 stretch: blank card set

Pick a food of your own (a can of soup, a bag of salad, a bottle of juice, a box of cereal). Cut out the ten blank cards below and fill them in for that food. Use the same ten step names. On each card, write one job, one guess at the time, and one guess at the cost share. Then answer:

- (a) If you could change one step to lower the price of your food, which step would it be?
- (b) Who would pay for that change instead?

1. GROW OR RAISE	2. HARVEST
3. PROCESS	4. PACKAGE
5. TRANSPORT	6. STORE
7. SELL	8. COOK
9. EAT	10. LEFTOVERS AND WASTE